debate, and it is far from clear what role it will have in the cosmological framework that will ultimately emerge from string theory. Achieving these cosmological insights will, no doubt, rely heavily on the ability of physicists to come to grips with all aspects of the second superstring revolution. What, for example, are the cosmological consequences of the existence of fundamental higher-dimensional branes? How do the cosmological properties we have discussed change if string theory happens to have a coupling constant whose value places us more toward the center of Figure 12.11 rather than in one of the peninsular regions? That is, what is the impact of full-fledged M-theory on the earliest moments of the universe? These central questions are now being studied vigorously. Already, one important insight has emerged.

M-Theory and the Merging of All Forces

In Figure 7.1 we showed how the strengths of the three nongravitational couplings merge together when the temperature of the universe is high enough. How does the strength of the gravitational force fit into this picture? Before the emergence of M-theory, string theorists were able to show that with the simplest of choices for the Calabi-Yau component of space, the gravitational force almost, but not quite, merges with the other three, as shown in Figure 14.2. String theorists found that the mismatch could be avoided by carefully molding the shape of the chosen Calabi-Yau, among other tricks of the trade, but such after-the-fact fine tuning always makes a physicist uncomfortable. Since no one currently knows how to predict the precise form of the Calabi-Yau dimensions, it seems dangerous to rely upon solutions to problems that hinge so delicately on the fine details of their shape.

Witten has shown, however, that the second superstring revolution provides a far more robust solution. By investigating how the strengths of the forces vary when the string coupling constant is not necessarily small, Witten found that the gravitational force curve can be gently nudged to merge with the other forces, as in Figure 14.2, without any special molding of the Calabi-Yau portion of space. Although it is far too early to tell, this may indicate that cosmological unity is more easily achieved by making use of the larger framework of M-theory.